

Bayesian Evidence Trees with Monte Carlo Uncertainty Propagation

Theory, Likelihood Estimation, Adversarial Analysis, and Implementation Guide

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Abstract

This document describes the theory, implementation, and practical application of Belief Engine (Python package: `bayes_tree`), a framework for evaluating complex hypotheses by decomposing them into hierarchical evidence trees. Top-level branches act as independent—or explicitly correlated—evidence combined at the root via likelihood ratios in log-odds space, while Monte Carlo simulation propagates uncertainty from likelihood ratio intervals to the posterior distribution. We present the formal model, provide systematic methods for estimating likelihood ratios from data, expert judgement, and the literature, and document the tool’s sensitivity analysis and importance ranking capabilities. We further introduce an *adversarial robustness analysis* framework that systematically challenges tree assumptions—prior specification, likelihood ratio calibration, distributional form, and evidence independence—to identify vulnerabilities and generate structured defenses.

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1 Introduction

Belief Engine is not an AI that thinks for you—it is a calculator for belief revision that forces you to make your assumptions explicit. The tool combines Bayesian inference with Monte Carlo simulation to propagate uncertainty through a tree of evidence. Each branch of the tree represents a piece of evidence characterized by a *likelihood ratio* (LR), and the tool computes a posterior probability distribution for the root hypothesis.

The motivation is to provide a structured, transparent framework for combining multiple heuristic likelihood ratios into a posterior probability. It is useful for exploratory “what-if” analysis, sensitivity studies, and producing an interpretable tree of how local evidence shifts beliefs. Evidence branches that share a common methodological flaw can be tagged with a `correlation_group` and correlation coefficient ρ , so that the engine models their dependence via a Gaussian copula rather than assuming independence.

2 Theoretical Background

2.1 Bayes’ Theorem

For a hypothesis H and evidence E :

$$P(H|E) = \frac{P(E|H) P(H)}{P(E)} \quad (1)$$

The **likelihood ratio** (LR) form is:

$$\text{LR} = \frac{P(E|H)}{P(E|\neg H)} \quad (2)$$

2.2 Log-Odds Representation

Converting probability to log-odds:

$$\text{lo}(p) = \ln \frac{p}{1-p} \quad (3)$$

And back:

$$p = \frac{1}{1 + e^{-\text{lo}}} \quad (4)$$

A Bayesian update in log-odds space is simply additive:

$$\text{lo}(P(H|E)) = \text{lo}(P(H)) + \ln(\text{LR}) \quad (5)$$

This additive property makes it natural to combine multiple independent pieces of evidence.

2.3 Combining Independent Evidence

Given n independent evidence branches with likelihood ratios $\text{LR}_1, \text{LR}_2, \dots, \text{LR}_n$, the combined posterior in log-odds is:

$$\text{lo}(\text{posterior}) = \text{lo}(\text{prior}) + \sum_{i=1}^n \ln(\text{LR}_i) \quad (6)$$

Equivalently, in odds form:

$$\text{odds}(H \mid E_{1:n}) = \text{odds}(H) \prod_{i=1}^n \text{LR}_i \quad (7)$$

The tool computes this by collecting all *leaf nodes* from the tree and summing their log-LRs against the root prior. Internal (non-leaf) nodes are pure groupers and carry no evidence. The conditional independence assumption applies to leaf nodes given H and given $\neg H$; see the warnings below.

2.4 Correlated Evidence (Correlation Groups)

When two or more branches share a common methodological flaw—for example, radiocarbon dates from the same laboratory with the same contamination source—the independence assumption is violated. The tool supports explicit modelling of such dependencies via `correlation_group` tags.

Branches sharing a group are sampled using a *Gaussian copula*:

1. Generate independent standard normals $Z_1, \dots, Z_k$ for the k grouped branches.
2. Correlate them: $Z'_j = \rho Z_1 + \sqrt{1 - \rho^2} Z_j$ for $j \geq 2$, where ρ is the user-specified correlation coefficient.
3. Transform to uniform quantiles via $U_i = \Phi(Z'_i)$.
4. Use U_i as the quantile input for each branch's LR sampler (inverse CDF).

The effect is that positively correlated branches tend to produce extreme LR values together, *widening* the posterior distribution and yielding more conservative credible intervals. Non-grouped branches remain independent.

2.5 Effective Likelihood Ratio

The *effective LR* is the single likelihood ratio that would produce the same posterior from the given prior:

$$\text{LR}_{\text{eff}} = \exp\left(\log(\text{posterior}) - \log(\text{prior})\right) \quad (8)$$

2.6 Monte Carlo Uncertainty Propagation

When the LR for each branch is uncertain (specified as an interval rather than a point value), the tool uses Monte Carlo simulation:

1. For each simulation run, sample an LR from each branch's distribution.
2. Compute the combined posterior using the log-odds sum.
3. Repeat N times (default: 10,000) to build a posterior distribution.
4. Report summary statistics: median, mean, 90% credible interval.

2.7 LR Sampling Distributions

Three distributions are available for sampling LR values from the interval `[lr_min, lr_max]`:

log_uniform (default) — Uniform in log-space. Best when uncertainty spans orders of magnitude. The sampled value is:

$$\text{LR} = \exp\left(\text{Uniform}(\ln(\text{lr_min}), \ln(\text{lr_max}))\right) \quad (9)$$

uniform — Uniform in linear space:

$$\text{LR} = \text{Uniform}(\text{lr_min}, \text{lr_max}) \quad (10)$$

beta — Beta distribution scaled to the interval, controlled by parameters `lr_alpha` and `lr_beta`:

$$\text{LR} = \text{lr_min} + \text{Beta}(\alpha, \beta) \times (\text{lr_max} - \text{lr_min}) \quad (11)$$

3 Tree Structure and Aggregation

The tree has two kinds of nodes:

- **Leaf nodes** carry evidence: each specifies a likelihood ratio or interval. Only leaves contribute to the posterior via the log-odds sum.
- **Internal nodes** are pure groupers. They must not carry `lr_min`, `lr_max`, or `likelihood_ratio`. The parser rejects any node that has both children and an LR parameter.

The root prior is the *only* prior. The posterior is computed as a flat traversal over all leaves:

$$\log(\text{posterior}) = \log(\text{prior}_{\text{root}}) + \sum_{\ell \in \text{leaves}} \ln(\text{LR}_{\ell}) \quad (12)$$

Internal nodes' contributions are computed for display only: the “subtree contribution” of an internal node is the combined LR of its descendant leaves, showing how each evidence group influences the overall posterior. These figures are informational and do not feed back into the computation.

Leave-one-out ranking. Removing an internal node removes all its descendant leaves from the sum. Removing a leaf removes just that leaf. The resulting change in the baseline posterior quantifies each node's importance.

4 Sensitivity Analysis

The tool provides two sensitivity measures:

1. **Threshold probabilities:** $P(\text{posterior} < 5\%)$, $P(\text{posterior} < 10\%)$, $P(\text{posterior} > 50\%)$.
2. **Leave-one-out ranking:** For each branch, compute the posterior without that branch. The difference from the baseline median shows how much each branch influences the result.

5 YAML File Structure

The input is a YAML file defining a tree of evidence. Below is the complete specification.

5.1 Root Node

```
node "Hypothesis_name"
prior 0.5 # Prior probability (0 to 1)
children
  - ... # List of evidence branches
```

5.2 Evidence Node

Each child node is either a **leaf** (carries an LR) or an **internal grouper** (has children but no LR).

Leaf node:

```
— node "Evidence_description"
  evidence_type for          # "for", "against", or "neutral"

  # Option A: Point estimate
  likelihood_ratio 2.5      # Exact LR value

  # Option B: Uncertainty interval (preferred)
  lr_min 0.01              # Lower bound of LR
  lr_max 0.10              # Upper bound of LR
  lr_dist log_uniform      # Distribution: log_uniform |
                           uniform | beta

  # Optional for beta distribution
  lr_alpha 2               # Beta shape parameter alpha
  lr_beta 2                # Beta shape parameter beta

  # Optional: correlation with other branches
  correlation_group c14    # Group tag (branches with same
                           tag are correlated)
  rho 0.7                  # Pearson correlation coefficient
                           (0 to 1)
```

Internal node (grouper):

```
— node "Evidence_category"
  evidence_type for          # For display only
  children                  # Leaf children carry the actual
                           evidence
    — node "Sub-evidence_A"
      lr_min 1.5
      lr_max 3.0
      evidence_type for
    — node "Counter-argument"
      lr_min 0.3
      lr_max 0.7
      evidence_type against
```

Rule: A node with children must not specify `lr_min`, `lr_max`, or `likelihood_ratio`. The engine raises a `ValueError` if this constraint is violated.

5.3 Evidence Type

for — Evidence supporting the hypothesis ($LR > 1$).

against — Evidence against the hypothesis ($LR < 1$).

neutral — Uninformative evidence ($LR \approx 1$).

The tool validates that `evidence_type` is consistent with the LR values and warns if there is a mismatch.

5.4 Likelihood Ratio Interpretation

LR value	Meaning
$LR \gg 1$	Strong evidence for the hypothesis
$LR > 1$	Moderate evidence for
$LR = 1$	Neutral (no update)
$LR < 1$	Evidence against
$LR \ll 1$	Strong evidence against

6 Estimating Likelihood Ratios

The quality of a Bayes Tree analysis depends on the quality of the likelihood ratio estimates assigned to each branch. This section presents systematic methods for arriving at defensible LR values, ranging from data-driven computation to structured expert judgement.

6.1 From Data: Frequency-Based Estimation

When empirical data are available, the likelihood ratio can be computed directly from conditional frequencies:

$$LR = \frac{P(E | H)}{P(E | \neg H)} = \frac{\text{rate of evidence } E \text{ when } H \text{ is true}}{\text{rate of evidence } E \text{ when } H \text{ is false}} \quad (13)$$

Example: diagnostic test. A blood test detects a disease in 95% of true cases (sensitivity = 0.95) and produces false positives in 5% of healthy individuals ($1 - \text{specificity} = 0.05$):

$$LR^+ = \frac{0.95}{0.05} = 19.0$$

A positive test result is strong evidence for disease ($LR \approx 19$). A negative result gives:

$$LR^- = \frac{1 - 0.95}{1 - 0.05} = \frac{0.05}{0.95} \approx 0.053$$

When exact frequencies are uncertain, use the range of plausible values as `lr_min` and `lr_max`. For instance, if sensitivity could be 90–98% and specificity 93–97%:

$$LR_{\min}^+ = \frac{0.90}{0.07} \approx 12.9, \quad LR_{\max}^+ = \frac{0.98}{0.03} \approx 32.7$$

6.2 From Published Studies

Many empirical fields report results from which LRs can be derived:

Relative risk (RR) or odds ratio (OR). When baseline rates are known, $LR \approx RR$ for rare outcomes. For case-control studies, $OR \approx LR$ when the sampling fraction is small.

Effect sizes. Cohen’s d can be converted to an approximate LR for observing a value above a threshold:

$$LR \approx \frac{\Phi(\text{threshold} - d)}{\Phi(\text{threshold})}$$

where Φ is the standard normal CDF. For a one-tailed observation at the median (threshold = 0): $LR = \Phi(-d/2)/\Phi(d/2)$.

Bayes factors. If a study reports a Bayes factor BF_{10} , this is *already* a likelihood ratio and can be used directly.

Confidence intervals. A 95% CI that excludes the null gives a rough lower bound on evidential strength. The further the CI is from the null, the stronger the evidence.

Handling uncertainty in published estimates. Use the study’s confidence interval or reported range to set `lr_min`/`lr_max`. If a meta-analysis reports $OR = 3.2$ with 95% CI $[1.8, 5.7]$, assign `lr_min` = 1.8, `lr_max` = 5.7.

6.3 From Expert Judgement: Structured Elicitation

When empirical data are sparse or unavailable, likelihood ratios must be elicited from domain expertise. The following structured methods reduce bias.

6.3.1 The Surprise Test

Ask: “How surprised would I be to see this evidence if H were true, compared to if H were false?”

Reaction if H true vs. false	Approximate LR	Strength
Equally likely either way	1	Neutral
Somewhat more expected if H	2–3	Weak
Clearly more expected if H	5–10	Moderate
Very surprising if $\neg H$	20–50	Strong
Nearly impossible if $\neg H$	100+	Decisive
Somewhat more expected if $\neg H$	0.3–0.5	Weak against
Clearly more expected if $\neg H$	0.1–0.2	Moderate against
Very surprising if H	0.02–0.05	Strong against

6.3.2 The Frequency Frame

Convert the question to a natural frequency to reduce cognitive bias:

“Out of 1000 cases where H is true, how many would show evidence E ? Out of 1000 cases where H is false, how many would show E ?”

If the answers are 800 and 200:

$$LR = \frac{800/1000}{200/1000} = 4.0$$

6.3.3 The Reference Class Method

Anchor the estimate to a comparable situation where data exist:

1. Identify a structurally similar domain where the same type of evidence has been studied empirically.
2. Use that domain’s LR as a starting point.
3. Adjust up or down for known differences between the reference and target.
4. Express residual uncertainty as `lr_min`/`lr_max`.

Example. You need the LR for “manuscript attributed to author X by stylometric analysis.” Stylometric authorship studies in English literature typically achieve LRs of 5–50 depending on method and corpus. If your case involves an unusually small corpus: `lr_min` = 2, `lr_max` = 15.

6.3.4 The Adversarial Test

For politically or emotionally charged hypotheses, adopt the strongest plausible version of the opposing view:

1. Estimate $\text{LR}_{\text{proponent}}$: the LR a reasonable proponent of H would assign.
2. Estimate $\text{LR}_{\text{sceptic}}$: the LR a reasonable sceptic would assign.
3. Use these as `lr_max` and `lr_min` respectively.

This ensures the interval spans the range of defensible interpretations.

6.4 Calibration Rules of Thumb

LR	Verbal label	Typical source
> 100	Decisive	Forensic DNA, mathematical proof
20–100	Strong	Controlled experiment, multiple witnesses
5–20	Moderate	Observational study, expert testimony
2–5	Weak	Anecdotal report, weak correlation
1	Neutral	Irrelevant information
0.2–0.5	Weak against	Minor disconfirmation
0.05–0.2	Moderate against	Failed prediction, conflicting data
0.01–0.05	Strong against	Robust negative result
< 0.01	Decisive against	Definitive refutation

6.5 Common Pitfalls

1. **Conflating LR with posterior.** An LR of 10 does *not* mean 90% probability. It shifts the odds by a factor of 10 — the posterior depends on the prior.
2. **Ignoring base rates.** When converting published statistics to LRs, ensure you are computing $P(E|H)/P(E|\neg H)$, not $P(H|E)$.
3. **Double-counting evidence.** If two branches draw on the same underlying data, they are not independent. Merge them or model one as a sub-child.
4. **Overconfidence in point estimates.** Unless the LR is known from large-sample data, prefer an interval. Specify `lr_min` and `lr_max` to represent your genuine uncertainty.
5. **Anchoring on round numbers.** Analysts tend to assign LRs of exactly 2, 5, or 10. Consider whether 3.5 or 7 might be more accurate.
6. **Symmetry bias.** Absence of evidence is *not* evidence of absence. If a search for evidence E was conducted and nothing was found, the LR depends on the search’s power to detect E if H were true.

6.6 Worked Example: Historical Hypothesis

Consider the hypothesis “Napoleon was deliberately poisoned with arsenic.”

Evidence: hair samples contain high arsenic.

- $P(E|H)$: If Napoleon was poisoned, high arsenic in hair is very likely. Estimate: 0.90–0.99.
- $P(E|\neg H)$: If not poisoned, Scheele's Green wallpaper, medicines, and environmental arsenic could still produce elevated levels. Studies show this was common in the 19th century. Estimate: 0.40–0.70.
- $LR = P(E|H)/P(E|\neg H)$: $lr_min = 0.90/0.70 \approx 1.3$, $lr_max = 0.99/0.40 \approx 2.5$.

The evidence is only weakly diagnostic because the alternative explanation (environmental arsenic) has high $P(E|\neg H)$.

Evidence: autopsy showed stomach cancer.

- $P(E|H)$: If poisoned, a stomach ulcer/tumour is possible but not expected. Estimate: 0.10–0.30.
- $P(E|\neg H)$: If not poisoned, stomach cancer explains the death directly. Estimate: 0.70–0.90.
- $LR = 0.10/0.90 \approx 0.11$ to $0.30/0.70 \approx 0.43$.

This is moderate evidence *against* poisoning.

7 Complete YAML Example

```
node "Patient_has_disease_X"
prior 0.10
children
  - node "Positive_blood_test"
    evidence_type for
    lr_min 3.0
    lr_max 8.0
    lr_dist log_uniform
    children
      - node "Test_specificity_concerns"
        evidence_type against
        lr_min 0.5
        lr_max 0.9

  - node "Family_history_present"
    evidence_type for
    likelihood_ratio 2.0

  - node "Age_below_typical_onset"
    evidence_type against
    lr_min 0.3
    lr_max 0.7
    lr_dist uniform

  - node "Symptom_pattern_matches"
    evidence_type for
    lr_min 1.5
```

```
lr_max 4.0
lr_dist beta
lr_alpha 2
lr_beta 3
```

8 Usage

8.1 Running the Tool

```
# Install
pip install bayes-tree

# Terminal UI with histogram and tree view
bayes-tree examples/mycase.yaml

# Machine-readable JSON output
bayes-tree examples/mycase.yaml --format json

# Prior sensitivity sweep
bayes-tree examples/mycase.yaml --prior-sweep

# Custom number of simulations
bayes-tree examples/mycase.yaml -n 50000

# Adversarial robustness audit
bayes-tree examples/mycase.yaml --adversarial

# Adversarial audit with JSON output
bayes-tree examples/mycase.yaml --adversarial --format json
```

8.2 Output Sections

The tool produces the following output:

1. **Validation warnings** — Flags any inconsistencies between `evidence_type` and LR values.
2. **Combined posterior** — Summary statistics (median, mean, CI) and ASCII histogram of the posterior distribution.
3. **Tree view** — Hierarchical display showing each branch's contribution, with prior → posterior transitions.
4. **Sensitivity analysis** — Threshold probabilities.
5. **Importance ranking** — Leave-one-out analysis showing which branches have the most influence.
6. **Adversarial audit** (with `--adversarial`) — Robustness report listing plausible attacks, severity ratings, and defense suggestions.

8.3 Tips for Creating YAML Files

1. **Start with the prior.** What is the base rate or initial probability before any evidence is considered?

2. **List independent evidence.** Each top-level child should be logically independent of the others.
3. **Prefer intervals over point estimates.** Using `lr_min` and `lr_max` captures your uncertainty about the strength of evidence.
4. **Use `log_uniform` for uncertain LRs.** When you know the LR is “somewhere between 0.01 and 0.1” but aren’t sure where, log-uniform gives appropriate weight to different orders of magnitude.
5. **Add children for drill-down.** Sub-nodes explain or decompose a parent’s evidence but don’t add to the root combination.
6. **Validate your intuitions.** Run the tool and check that the output matches your qualitative expectation. If not, reconsider your LR estimates.

9 Adversarial Robustness Analysis

A Bayesian evidence tree encodes several assumptions: the prior probability, the likelihood ratio estimates, the distributional forms, and the independence of top-level branches. An *adversarial audit* systematically challenges each of these assumptions with plausible perturbations, measures the impact on the posterior, and reports which assumptions the conclusion is most sensitive to. This mirrors the role of a sceptical peer reviewer who searches for the weakest points in an argument.

9.1 Motivation

Even a well-constructed evidence tree may be fragile: the conclusion may depend critically on a single uncertain LR estimate, or collapse if two branches are correlated rather than independent. Standard sensitivity analysis (Section 4) varies one parameter at a time; adversarial analysis goes further by (i) testing specific *types* of methodological challenge, (ii) scoring each attack’s real-world *plausibility*, and (iii) searching for *minimal attack combinations* that flip the conclusion.

9.2 Attack Types

The framework implements four attack classes, each targeting a different modelling assumption.

9.2.1 Prior Bias Attack

The prior $P(H)$ is often the most contested parameter. The prior bias attack tests robustness by sweeping the prior through a range of values $\{0.01, 0.05, 0.10, \dots, 0.99\}$ and reports:

- The posterior at each perturbed prior.
- The *flip prior* p^* : the threshold prior above (or below) which the posterior crosses 50%, found by binary search:

$$p^* = \arg \min_p |\tilde{P}(H | E; p) - 0.50|$$

where $\tilde{P}$ denotes the median posterior from Monte Carlo simulation with prior p .

9.2.2 LR Calibration Attack

This attack questions the likelihood ratio estimates by applying two perturbations:

1. **Shrinkage toward neutrality.** The log-LR is multiplied by a shrinkage factor $\gamma \in \{0.25, 0.50, 0.75\}$:

$$\log \text{LR}' = \gamma \cdot \log \text{LR}$$

This pulls extreme LR values (both strong support and strong counter-evidence) toward the neutral value $\text{LR} = 1$.

2. **Uncertainty expansion.** The log-space interval $[\log \text{LR}_{\min}, \log \text{LR}_{\max}]$ is expanded by a factor $\lambda \in \{1.5, 2.0, 3.0\}$ around the centre:

$$\log \text{LR}'_{\min/\max} = c \mp \lambda \cdot \frac{w}{2}, \quad c = \frac{\log \text{LR}_{\min} + \log \text{LR}_{\max}}{2}, \quad w = \log \text{LR}_{\max} - \log \text{LR}_{\min}$$

9.2.3 Distribution Misspecification Attack

The choice of sampling distribution (log-uniform, uniform, or beta) affects the posterior. This attack tests the sensitivity to that choice by replacing each branch's distribution with alternatives:

- Log-uniform $\rightarrow$ uniform, $\text{beta}(\alpha=2, \beta=5)$, $\text{beta}(\alpha=5, \beta=2)$
- Uniform $\rightarrow$ log-uniform, beta variants

If the posterior is stable across distributional assumptions, the conclusion is more trustworthy.

9.2.4 Correlation Attack

The standard model assumes independence of top-level branches. The correlation attack introduces dependence between a pair of branches using a Gaussian copula:

1. Draw correlated standard normals (Z_1, Z_2) with correlation ρ :

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$$

2. Transform to uniform quantiles via $U_i = \Phi(Z_i)$.
3. Use U_i as the quantile input for each branch's LR sampler.

This is tested for $\rho \in \{0.3, 0.5, 0.7, 0.9\}$. The attack is most plausible when two branches share a common information source.

9.3 Plausibility Scoring

Not every mathematically possible attack is realistic. Each attack receives a *plausibility score* on a 0–10 scale, based on:

- **Evidence type.** Qualitative evidence (testimony, motive, circumstantial claims) is more attackable than quantitative evidence (DNA, isotope analysis, controlled experiments).
- **LR extremity.** Very large LR values (> 100) are harder to justify and thus more attackable.
- **Correlation context.** Same-source evidence pairs receive higher correlation plausibility.
- **Prior shift magnitude.** Small prior shifts are more plausible than extreme ones.

Only attacks with plausibility ≥ 3.0 (configurable) are reported.

9.4 Attack Search

Given an *attack budget* k (default $k = 3$, modelling a realistic peer review), the search algorithm finds the minimal combination of attacks that flips the conclusion below a threshold (default 50%).

The greedy algorithm proceeds as follows:

1. Score each plausible attack by $\text{impact} \times (0.5 + 0.5 \cdot \text{plausibility}/10)$.
2. Select the highest-scoring attack, apply it to the tree.
3. Re-evaluate remaining attacks on the modified tree.
4. Repeat up to k times or until the conclusion flips.

9.5 Severity Classification

Each attack result is classified by the absolute change in posterior median:

Severity	$ \Delta_{\text{posterior}} $	Interpretation
Critical	$> 20\%$	Conclusion may not survive scrutiny
High	$> 10\%$	Significant vulnerability
Moderate	$> 5\%$	Non-trivial sensitivity
Low	$\leq 5\%$	Minor impact

9.6 Defense Generation

For each successful attack, the system generates structured defenses:

- **Prior bias:** Justify the prior with base rates; argue why priors above the flip threshold are unreasonable.
- **LR calibration:** Cite specific studies supporting the LR estimate; compute LR from documented base rates.
- **Misspecification:** Justify the chosen distribution; show robustness by running with all three distributions.
- **Correlation:** Add corroborating independent evidence; document source independence explicitly.

9.7 Interpreting the Audit

The audit produces a verdict:

- **ROBUST:** No high-severity attacks found; the conclusion withstands plausible challenges.
- **CAUTIOUS:** High-severity attacks exist but do not flip the conclusion; additional evidence or justification is recommended.
- **VULNERABLE:** The conclusion can be flipped by a plausible combination of attacks; the argument requires strengthening.

10 Limitations and Assumptions

- **Independence assumption.** Leaf nodes are assumed conditionally independent given H and $\neg H$ by default. Sibling leaves that are facets of the same underlying observation (e.g., three laboratories running C14 on samples from the same cloth) violate this assumption. Mitigations: (a) merge such leaves into one; (b) keep only the strongest; or (c) use `correlation_group` and `rho` to model the dependence explicitly. The adversarial correlation attack (Section 9.7) can quantify the impact of any remaining unmodeled correlations.
- **Leaves-only evidence.** Only leaf nodes carry likelihood ratios; internal nodes are pure groupers. A YAML file with an LR on a node that also has children is rejected at parse time. This design eliminates double-counting that arises from evidence appearing both at the parent and child level.
- **Log-uniform bias.** The default log-uniform sampling biases toward multiplicative scales. Users should choose a distribution that reflects their actual beliefs about where in the interval the true LR lies.
- **Subjective LR estimates.** The tool is only as good as the likelihood ratios you assign. It makes subjective reasoning transparent and quantitative, but does not eliminate subjectivity.
- **Numerical stability.** The implementation clamps probabilities and log-odds to safe numerical ranges to avoid overflow, which may slightly affect extreme posteriors (very close to 0 or 1).

11 Implementation Notes

- **Log-odds stability:** Probabilities are clamped to $[10^{-12}, 1 - 10^{-12}]$ before log-odds conversion; log-odds are clamped to $[-700, 700]$ before exponentiation.
- **Sampling:** The same random draw strategy is applied per simulation; users can set simulation counts via the command line.
- **Tree view:** A recursive collection routine computes subtree-contribution statistics for internal nodes (their descendant leaves' combined LR and hypothetical posterior) purely for display. The root prior is the only prior used; internal nodes do not have independent posteriors.
- **Effective LR for root:** The root node's LR interval is derived from the 5th–95th percentiles of the effective LR distribution across all Monte Carlo runs.

12 Dependencies

The core engine requires:

- Python 3.9+
- PyYAML (`pip install pyyaml`)

Optional dependencies for visualisation: `matplotlib`, `numpy`. Optional for PDF reports: `reportlab`.